

Daniel O. Seguin

Legacy Maintenance, Modernization & Technical Documentation Specialist (AI-Accelerated)

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PROFILE

IT Analyst/Developer with 30+ years of experience specializing in legacy application maintenance, system documentation, and risk-free modernization. Expert in extending mission-critical systems while guiding organizations toward scalable, modern architectures with AI-enhanced development practices. Hands-on background across Informix 4GL, Genero, PowerBuilder, Classic ASP, COBOL, and RPG, plus major databases including Informix, SQL Server, Oracle, MySQL, Dataverse, and PostgreSQL.

CORE STRENGTHS

- Legacy System Expertise: Support, enhance, and troubleshoot complex enterprise systems amid talent shortages.
- Modernization Without Disruption: Reverse-engineering and documentation (DbDocGen, ERD Generator) of multi-decade databases.
- API & Platform Modernization: PowerBuilder 2025 → C# .NET API services with auto-generated Swagger and VS-ready projects.
- AI-Assisted Development: Developing solutions integrating generative AI, developed modern workflows using GitHub Actions to accelerate delivery and reduce defects.
- Architecture: Developed service-oriented architecture either using API Services or JSON or XML data files and messages.
- Cloud-Ready Migration: Transitioning legacy systems to Azure. Power Platform, and modern database environments.

SKILLS & TECHNOLOGIES

- Languages & Frameworks: Informix 4GL, PowerBuilder, COBOL, RPG, C#, Java, Python, PowerShell, SQL, PL/SQL, .NET 6/8, TypeScript, Angular.
- Databases: Informix, SQL Server, Oracle, MySQL, PostgreSQL, Dataverse.
- Tools & Platforms: PowerBuilder 2025, GitHub Copilot, GitHub Actions, Microsoft 365 Copilot, PowerApps, PowerAutomate, Power BI, Jira, Confluence, SSRS, SAP BusinessObjects.
- Cloud & DevOps: Azure, Azure DevOps, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions), MCP Servers, FastMCP.cloud.
- Operating Systems: Windows, Linux, Unix.

PROFESSIONAL EXPERIENCE

Data Analyst / Web Developer / Full-Stack Developer (in collaboration with GitHub Copilot Agent)

September 2024 – Present

The following projects are internal, or client projects intended to work with new technologies.

- **Internal Project: Consultant Companion**

Consultant Companion is a comprehensive business management application developed with PowerBuilder 2025 and integrated with Power Automate and PowerApps, designed specifically for consultants and small consulting firms. It centralizes and streamlines operations across multiple business domains.

The solution was developed using Dynamics 365, Power Tools, and PowerBuilder 2025. It enables consultants to manage core business entities such as clients, engagements, employees, expenses, timesheets, invoices, projects, and accounting (including general journal entries and chart of accounts).

The application also integrates with Sage Accounting for financial operations, ADP for payroll, and Microsoft Power Platform components (Dataverse, model-driven apps, Power Pages, Power Automate, and Power BI).

- **Internal Project: DbDocGen**

DbDocGen (2025) is a Python-based solution designed for reverse engineering various databases (SQL Server, Oracle, Informix, PostgreSQL, Dataverse/PowerApps) to produce data dictionaries in Markdown, PDF, or HTML formats.

The data dictionary can also be used to generate ERD diagrams viewable in Draw.io and Lucidchart.

- **Internal Project: DBA Helper**

Development of a DBA tool for monitoring database server statistics.

This application was developed in Python.

- **Client Projects: Responsive Website Development and Maintenance**

Development and maintenance of responsive websites adapted to multiple device types and compliant with accessibility standards for visually impaired users:

- Roxane Bélanger, Speech-Language Pathologist — <https://roxanebelanger.com>
Migration from .NET to Angular
- SeguinDev Corporate Website — <https://daniel.seguin.dev>
- Jessica Larmour, Real Estate Leasing Specialist — <https://montreal4rent.com>
Integrated with the Web Companion application
- Web Companion — A real estate management web application exchanging data via JSON files and integrated with Google Drive:
<https://realestatewebcompanion.seguin.dev>

- **Internal Project: Data Validator**
Development of a Python-based data validation tool to help data analysts perform rigorous validation of database tables and associated business rules.
- **Internal Project: DemoStore**
Development of a complete e-commerce web application demonstrating secure, PCI DSS-compliant payment integration using Stripe Elements and the PaymentIntents API. The application includes a session-based shopping cart, real-time webhook event handling, and a REST API (Django REST Framework + *drf-spectacular*) connected to Microsoft SQL Server.
Credit card data never transits through the application server—Stripe.js sends it directly to Stripe, eliminating any server-side PCI exposure.
- **Internal Project: Legacy Application Migration (PowerBuilder → Django/Python)**
Modernization of an aging desktop application originally developed in PowerBuilder, rebuilt as a full web application using Python and the Django Framework. The legacy system managed core operations of a consulting firm—tracking consultants, clients, projects, timesheets, and expenses—but relied on outdated proprietary technology that was difficult to maintain.
The rewritten application delivers the same functionality through a modern browser-based interface, with a codebase that is easier to maintain, scale, and deploy.
- **Internal Project: MyAccounting — Desktop Accounting Application**
Desktop accounting application developed in Python (tkinter) for small Canadian service businesses. Designed as an alternative to Sage 50 Accounting Pro, it enables migration to an in-house solution and saves nearly \$1,000 per year in recurring license fees. The application manages the full accounts receivable lifecycle: invoicing with PDF generation, payments, general ledger, accounting journals, and financial statements (balance sheet, income statement, trial balance).
It includes a real-time dashboard, role-based access control, flat-file data imports, and Excel exports, and connects to Microsoft SQL Server via ODBC.

Public Works Canada — Analyst Developer

September 2024 – Present

- Provide second-line application support and ongoing development for DISIS and PSDCA systems (Sybase ASE, SQL Server).
- Troubleshoot and resolve production issues using PowerBuilder debugger and database tools; perform root-cause analysis and fixes.
- Manage Agile tasks and issues using Jira and Devops; communicate findings and recommendations to management.
- Perform QA reviews and support technical architecture and application security processes.
- Migrations: Java Mail app from Oracle Java to Microsoft JDK as a Windows Service; PowerBuilder apps (2017 → 2025); source code from SVN → Git (Bitbucket); DB connectivity from Sybase ASE → SQL Server.

- Developed PowerShell tool to migrate Jira tickets to Azure DevOps with Power BI dashboards (DAX) for progress tracking.
- Authored Java utility to encrypt credentials in configuration files and a SQL Server data model dictionary.
- Built Python utilities: data extractors to CSV/Excel; PowerBuilder source search; data flow diagram generator; Sybase→SQL Server migration validation.
- Created C# .NET API services for DISIS and PSDCA via PowerBuilder 2025 API generator.
- Modified PSDCA XML exchange processes with third parties (OLISS, RCMP, CSIS).
- Accelerated SQL migration and code validation (Java, PowerBuilder, PowerShell, SQL, Python).
- Deployed MSI Packages using Advanced Installer Application to deploy DISIS and PSDCA PowerBuilder applications with Spif release documents requested by PSPC package team to send to Shared Services Canada.
- Implemented the Devops legacy applications dashboard to do daily support and maintenance, to manage backlogs via parking logs, and to manage releases.
- Developed Devops queries to do queries used in the dashboard, and to do work item reporting.
- Created documentation in Devops wiki, created Lucidchart architecture, application, and database ERD diagrams.
- Developed a Python program that acts as report scheduler for DISIS and PSDCA PowerBuilder applications.
- Developed PowerShell and Python programs to extract database meta data to generate data model and ERD diagrams in Drawio which can be imported into Lucidchart.
- Modernize PowerBuilder applications DISIS and PSDCA to use modern authentication either by database login, LDAP login, or SSO.
- Prepared DISIS and PSDCA projects to start using agile ceremonies like sprint planning, scrums, sprint planning, and sprint retrospective.
- Analyzing, documenting, and planning how to replace the use of network drives as SSC will be retiring their use as of October 2026.

Ministry of Natural Resources of Quebec — Analyst Developer

December 2023 – September 2024

- Maintained Registre Foncier du Québec (PowerBuilder 2019–2022) using PFC libraries.
- Supported Solaris-to-Linux migration by preparing and executing test plans.
- Diagnosed and resolved production issues across Oracle PL/SQL, ETL, Korn Shell, COBOL, Java, and PowerBuilder components.
- Built complex SQL queries to support test execution and assist non-technical analysts.
- Migrated Unix Korn Shell scripts to Windows PowerShell.
- Conducted web accessibility and mobility testing for Geo-Immobilier (Angular/JavaScript).

- Documented technical work in Confluence; followed DMR methodology; used Jira (Scrum/Kanban).

Ministry of Environment of Quebec — Analyst Developer

June 2023 – November 2023

- Resolved production issues in Sago using PowerBuilder 2021 debugger and database tools; performed root-cause analysis and alternatives. Sago application was developed with PFC libraries.
- Documented technical procedures, maintained and enhanced application programs.
- Supported DevOps operations (dashboards, Git management, test plan maintenance).
- Designed ETL strategy to migrate data from Sago to external CRM (Logient) using PowerBuilder and PL/SQL.

Skechers — Analyst Developer

August 2022 – June 2023

- Analyzed business requirements and enhancements for Garpac (Order & Warehouse Management).
- Developed and maintained applications using Informix 4GL and Genero Desktop Client.
- Supported Wholesale (ERP), Retail, and E-Commerce operations.
- Resolved production issues using Informix 4GL Debugger, Genero Studio, and DB tools.
- Migrated Genero programs to newer versions; configured client/server environment.
- Created and maintained ETL scripts for data transformation within Garpac.
- Authored functional and technical documentation using Confluence and Lucidchart; managed requests via ServiceNow.
- Led PoC to migrate Informix 4GL source control from RCS to GitHub.
- Documented BPM for integrating order entry, warehouse, and Salesforce.
- Usage of ServiceNow ticket system to manage issues from beginning to end; added knowledge base documents.

Industrial Alliance — Analyst Developer

June 2021 – August 2022

- Designed data migration process, documented database entities and metadata in Markdown.
- Created ETL scripts to reverse engineer Equisoft Connect DB using CSV files, staged DB for cleansing.
- Coached junior analysts and developers (Markdown, VS Code, SQL Server, SSMS, Git, DevOps).

- Developed components in Azure Data Factory (linked services, pipelines, datasets, dataflows).
- Built PowerShell, C#, and SQL scripts; managed work in Scrum (ADO).
- Modeled SQL Server and JSON data (PowerDesigner); modeled Salesforce (ERBuilder).
- Developed ADF pipelines; performed integration and performance testing to load data into Salesforce.
- Configured Salesforce custom fields; built queries in Salesforce Workbench.
- Analyzed and documented applications for decommissioning; supported required business apps; participated in TFS testing.
- Usage of ServiceNow ticket system to manage issues from beginning to end; added knowledge base documents.

Desjardins — Functional Analyst

March 2021 – June 2021

- Modeled the current notary deposit process for automation aligned with business requirements.
- Modeled documents and email injection processes for improved efficiency.
- Collaborated in Scrum ceremonies and with business analysts.

SeguinDev

Various Projects — Analyst Developer

July 2020 – March 2021

- Synapsia (Neuligent): Led interim dev team; prepared user stories; built Scrum boards in ADO; developed backend scripts (Node.js, NPM, JavaScript); performed system/integration testing; documented and supported pre-production.
- RAD Exploration: Studied market niches; developed apps using Genero architecture.
- EFICIO (Integra-co): Maintained Windows Server 2016 infrastructure monthly.
- Four JS Tools: Created REST APIs using Genero/Informix 4GL; tested with Postman/Talend; performed code reviews and documentation.
- Paradox Security Systems: Built Python reporting programs integrated with SQL Server.
- System Administration: Managed Office365, SharePoint, Teams for condo and family organizations.
- Physiogougeon Web App: Developed/maintained C# MVC app with EF; administered VPS hosting and SQL Server; resolved production issues and performed debugging.

Air Canada Vacations — Analyst Developer

December 2019 – July 2020

- Analyzed business requirements for report migration from WebFocus to SAP BusinessObjects.
- Developed Web Intelligence reports using SAP BO Universe (BI).

- Managed SAP BO CMC (user security, universe management, report deployment, version control).
- Provided pre-production support and troubleshooting for migrated reports.
- Maintained and enhanced ERP Back-Office programs (Informix Genero 4GL) integrating with external systems via APIs.
- Maintained SSIS processes for importing vacation orders; documented ERP processes in Confluence; worked with Jira (Kanban).

Ministry of Justice — Functional Analyst

July 2019 – September 2019

- Created and executed test plans during ERP migration from Dynamics 2011 to 365.
- Participated in Scrum ceremonies, sessions with business analysts.
- Used DevOps for ticketing; created test plans (ADO/TFS).

EFICIO — Analyst Developer

June 2018 – July 2019

- Project 10a (Bizagi – OPEQ): Designed, developed, and maintained HR BPM workflow system using Bizagi; gathered requirements; authored specs; delivered in Agile sprints; improved release management; provided production support and troubleshooting (Bizagi Studio, SQL Server, JavaScript); executed test plans; configured environments on Windows Server 2016; created SSRS reports; mentored colleagues on Agile and release management.
- Project 10b (BI – IPEX): Developed and maintained SSRS reports and complex SQL for MES Hydra MOC; analyzed business requirements for OEE reporting; acted as technical analyst for SAP Concur; introduced/configured DevOps in Scrum; provided production support and RCA using Visual Studio and SQL Server.

Alithya — Analyst Developer

November 2017 – May 2018

- Collaborated with functional analysts to gather requirements and translate them into technical solutions.
- Created detailed organic analysis to minimize impacts and stay on budget.
- Wrote and executed test cases; participated in daily scrums; performed Agile tasks in Jira; maintained and developed solutions in an order management app (CLIC) using Velocity, JavaScript, XML, XSLT; used Jira for ticketing.

General Dynamics Information Technology — Analyst Developer

March 2016 – October 2017 (continued part-time support Nov 2017 – Oct 2018)

- Supported production environment using AnyDoc OCR, improved performance and stability.
- Troubleshot business issues across SSMS, VB6, AnyDoc, MS Access, VBA (Excel/Outlook), Visual Studio Debugger.

- Designed/implemented wire payment remittance processing solution integrating email processing, OCR, SQL Server, and C# MVC4.
- Gathered requirements, authored functional documents, translated into Agile user stories.
- Developed/maintained MVC4 apps in C# for operations and asset management.
- Configured Azure DevOps (TFS) for Agile PM: ticketing, team collaboration.
- Created SSRS dashboards, SSIS packages, ETL scripts for document processing/reporting; administered SQL Server and Access; built stored procedures, agent jobs, shell scripts.
- Maintained AnyDoc form families and Exchange IT jobs; programmed in VBScript, batch, VBA, and VB6.
- Deployed via IIS; configured Jaspersoft for self-service reporting; reverse-engineered and modeled DBs using PowerDesigner.
- Provided technical support during MBI upgrade testing, supported help desk, network, and printer issues.

Public Works of Canada — Analyst Developer

June 2015 – February 2016

- Supported users, developers, and testers with SPMIS reporting in prod and test environments.
- Troubleshoot PowerBuilder and SQL Server issues; provided immediate fixes and RCA.
- Gathered requirements; developed specs, estimates; maintained functional specs and report models.
- Investigated the SPMIS PowerBuilder 12.5 applications using debugger which was using PFC libraries.
- Created Excel tools with VBA for specs and test plans.
- Developed VB.NET tool to reverse engineer SSRS Report Models into SQL Server to generate documentation.
- Updated SPMIS Help file using MS Help Template and SVN; prepared functional specs for SSRS.

PowerStream — Analyst

February 2015 – May 2015

- Documented and executed applications upgrade deployment tasks using I-TOP across environments.
- Managed and version-controlled Java and COBOL source code, prepared and packaged releases.
- Generated Javadoc and Application Viewer documentation for each release; reviewed SQL scripts.

Public Works of Canada — Analyst Developer

October 2013 – February 2015

- Technical analysis and production support for SMIS (incident reporting): PowerBuilder app, Oracle DB, Cognos reports.
- Diagnosed/resolved production issues; RCA; proposed alternatives.
- Designed/developed enhancements to AFDMS integrating with SAP Financial.
- Authored documentation: variance/impact analysis; technical specs; collaborated with cross-functional teams.
- Developed AFDMS–SAP integration (PowerBuilder 12.5, SQL/PLSQL).
- Investigated the SMIS and AFDMS PowerBuilder 12.5 applications using debugger which was using PFC libraries.
- Created/executed test plans; implemented dashboard to track Agile testing progress and issues.
- Developed Oracle stored procedures to validate SAP data.

ERTH Corporation — Analyst Developer

December 2010 – September 2013

- Technologies: Informix, SQL Server, Java, Unix, Linux, Microsoft Dynamics Great Plains, PowerDesigner, Iron Speed Designer, .NET, Classic ASP, VBScript, JavaScript.

Seguin Consulting Services (Various Clients) Analyst Developer

April 2002 – April 2004

- Technologies: PHP, Classic ASP, MySQL, Unix, Business Vision, Acomba.

Nortel Network — Analyst Developer

October 2000 – March 2002

- Technologies: Informix 4GL, Informix SQL, Unix, PowerBuilder, MS-Access, PowerDesigner.

Harris Computer Systems — Analyst Developer

December 1991 – December 2010

- Technologies: Informix 4GL, Informix SQL, SQL Server, Oracle, PL/SQL, Classic ASP, VBScript, Java Spring MVC, JavaScript, PowerBuilder, PowerDesigner, VB6, Crystal Reports.

Domtar — Junior Developer (Young Professional Program)

September 1988 – November 1991

- Assignments in Goderich (Sifto), Donnacona (Paper Mill), and Montreal (Head Office, paper mill division).
- Technologies: dBASE IV, FoxPro, Informix 4GL and SQL.

PASSION PROJECTS & TECHNICAL EXPLORATION**Exploration with Power Tools, Azure, CI/CD, Docker, Kubernetes**

- Prototyped integrations between Power Tools, PowerBuilder, and Azure SQL Server.
- Leveraged GitHub Copilot to accelerate SQL migration from Sybase to SQL Server and assist in multi-language development (Java, PowerBuilder, PowerShell, .NET).
- Used GitHub Copilot to validate source code across various languages including SQL, Java, .NET, PowerBuilder, COBOL, and PowerShell, within both Visual Studio and Visual Studio Code.
- Utilized Microsoft 365 Copilot for both technical and non-technical tasks as a complement to GitHub Copilot.
- Developed Terraform scripts in PowerShell to support Azure resource development, validation, and deployment.
- Configured Windows and Linux environments using Docker and Kubernetes for development projects.
- Explored Microsoft Azure Free Tier and Pay-As-You-Go services for experimentation and testing.

- Set up CI/CD pipelines for .NET Dynaweb applications and legacy PowerBuilder applications using batch files and Python scripts.
- Developed sample web application with ASP.NET Core Identification Authentication using Blazor framework.
- LinkedIn training on Kafka and created samples programs.

Exploration building AI Solutions

- Learning the development of Model Context Protocol (MCP) servers and deployment to the cloud on FastMCP.cloud
- Learning the planning and development of AI Solutions using Anthropic foundation and framework.
- Developing client and server programs using Python with various libraries.

EDUCATION & CLEARANCES

- B.Sc. in Computer Science — University of Ottawa (1988).
- Security Clearance — Enhanced: File No. 95-60-7790-0003180084 (Expires 2034-05-30).
- Security Clearance — Secret: File No. 95607790-0003275929 (Expires 2035-02-21).